

Designing a potentiostat for differential pulse voltammetry

by

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> *A very long quote*
> *from someone very important.*
> – Mr M

Thank you to Leanne for helping to prepare this template.

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Abstract

Uittreksel

Table of Contents

Abstract **iii**

Uittreksel **iv**

Table of contents **v**

List of figures **vii**

List of tables **vii**

Nomenclature **ix**

Acronyms and abbreviations x

Definitions xi

1. Introduction **1**

1.1. Background and Motivation 1

1.2. Problem Statement 1

1.3. Project Objectives and Scope 2

1.4. Report Layout 2

2. Literature **3**

2.1. Introduction to review 3

2.2. Electrochemical background 3

2.2.1. Basics of electrochemical cells 3

2.2.2. Voltammetry as a family of techniques 4

2.2.3. Differential pulse voltammetry theory 5

2.2.4. Waveform description 5

2.2.5. Why DPV improves sensitivity 6

2.2.6. Typical analytical applications 6

2.3. Potentiostat fundamentals 7

2.3.1. The role of a potentiostat 7

2.3.2. The classic 3-op-amp potentiostat topology 7

2.3.3. Key performance parameters 8

2.3.4. Analog front-end approaches . . . 9
2.4. Survey of existing potentiostat designs . . . 10
2.4.1. Commercial instruments . . . 10
2.4.2. Early open-source designs . . . 10
2.4.3. General-purpose / high-performance open-source designs . . . 11
2.4.4. Portable / wireless / smartphone-integrated designs . . . 11
2.4.5. EIS-capable / AFE-chip-based designs . . . 11
2.4.6. Self-hosted web-interface / onboard-display designs . . . 12
2.4.7. Machine-learning-assisted voltammetry . . . 12

3. System Modeling & Theoretical Design 14
3.1. System Architecture . . . 14
3.2. Electrochemical Cell & Equivalent Circuit Modeling . . . 15
3.2.1. Proving Capacitive Current Rejection in DPV . . . 16
3.2.1.1. Total Current at the Electrode . . . 16
3.2.1.2. The Pulse-Sampling Scheme . . . 18
3.2.1.3. The Differential Proof . . . 18
3.2.1.4. Why the Output Forms a Peak . . . 18
3.3. Analog Front-End Stability & Control Loop Modeling . . . 19
3.3.1. Impact of C_{dl} on TIA Loop Stability . . . 19
3.3.1.1. Circuit Model and Node Equation . . . 19
3.3.1.2. Noise Gain and the Loop Gain Pole . . . 20
3.3.1.3. Phase Margin Degradation . . . 20
3.3.1.4. Compensation . . . 21
3.4. Voltammetric Digital Signal Processing & Peak Extraction . . . 21
3.5. Embedded Communication Protocol & GUI State Machine . . . 22

4. Detailed Hardware & Software Implementation 23
4.1. Choosing a microcontroller . . . 23
4.1.1. Choosing the microcontroller family . . . 23
4.2. Physical Circuitry & Electrode Interface Design . . . 24
4.3. Analog Front-End (AFE) Register Configuration . . . 24
4.4. ESP32-S3 Firmware Architecture & Real-Time Loop . . . 25
4.5. Wireless Web GUI & Interface Realization . . . 25

5. Summary and conclusions 26

References 27

A. Appendix title goes here 30

List of Figures

2.1. The three-electrode electrochemical cell configuration (working, reference and counter electrodes). 4
2.2. The DPV excitation waveform, showing the staircase ramp, superimposed pulses, and the two current-sampling instants (I_{before} and I_{end}) per step. . . 6
2.3. The classic 3-operational-amplifier potentiostat topology, showing the control amplifier, RE buffer, and TIA connected to the three electrodes. 8

3.1. System Architecture Diagram 15
3.2. Randles equivalent circuit modeling the electrochemical cell interface. 16
3.3. Finite state machine diagram for managing asynchronous tasks and WebSocket communication. 22

4.1. Custom two-layer PCB layout highlighting ground plane stitching and analog/digital signal separation. 24
4.2. Real-time web interface served from SPIFFS, plotting DPV scans via WebSockets. 25

List of Tables

2.1. Comparison of discrete op-amp and integrated AFE IC implementation approaches. 10
2.2. Comparison of surveyed potentiostat designs against the proposed architecture. 13

3.1. Comparison of Capacitive and Faradaic Currents 17

4.1. Comparison of candidate microcontroller families. 24
4.2. Comparison of candidate ESP32 variants. 24

Nomenclature

Acronyms and Abbreviations

ADC	Analog-to-Digital Converter
AFE	Analog Front-End
BLE	Bluetooth Low Energy
BOM	Bill of Materials
CA	Chronoamperometry
CE	Counter Electrode
CV	Cyclic Voltammetry
DAC	Digital-to-Analog Converter
DPV	Differential Pulse Voltammetry
EIS	Electrochemical Impedance Spectroscopy
FreeRTOS	Free Real-Time Operating System
GUI	Graphical User Interface
IoT	Internet of Things
LOD	Limit of Detection
LSV	Linear Sweep Voltammetry
MCU	Microcontroller Unit
MFT	My Funky Thesis
ML	Machine Learning
RE	Reference Electrode
SNR	Signal-to-Noise Ratio
SoC	System-on-Chip
SPE	Screen-Printed Electrode
SWV	Square Wave Voltammetry
TIA	Transimpedance Amplifier
WE	Working Electrode

Definitions

S

Student is an entity needing a thesis to transcend the state of being a student.

Chapter 1

Introduction

1.1 Background and Motivation

The global landscape of analytical chemistry is currently experiencing a significant shift towards point-of-care medical diagnostics and on-site environmental testing. In South Africa, a critical public health and environmental challenge is the accumulation of active pharmaceutical ingredients, specifically antiretroviral medication, in local water systems. Monitoring these specific pharmaceutical compounds at trace levels requires highly sensitive analytical techniques. Differential Pulse Voltammetry (DPV) was selected as a technique to detect these pharmaceutical compounds in the water due to its high sensitivity. DPV superimposes small pulses of potential onto a linear potential ramp or staircase. By measuring the current difference before and near the end of each pulse, the technique reduces background charging current and achieves high sensitivity. This makes DPV a foundational technique for isolating the minute faradaic currents required to detect trace pharmaceuticals outside of a laboratory environment.

1.2 Problem Statement

The primary barrier when testing water in South Africa is challening because of the tools required to perform accurate voltammetric analysis are costly, complex and lack in portability. The current commerically available potentiostat, such as the Palmsens4, systems serve as the gold standard for electrochemical analysis. However, these instruments are cost-prohibitive, physically bulky, and operate as closed-source devices. Consequently, it is impossible to integrate them deeply into custom, low-cost microfluidic or Internet of Things (IoT) architectures. While various open-source alternatives have been developed over the years, many remain tethered to external computers for power and data processing, or rely on proprietary applications that can become obsolete. This presents a distinct

need for a smart, autonomous, wireless, and low-cost potentiostat alternative capable of conducting highly sensitive DPV measurements independently in the field.

1.3 Project Objectives and Scope

The primary objective of this project is to design and develop an open-source, wireless potentiostat tailored specifically for the DPV detection of antiretroviral medications in water. The scope of this project encompasses the following hardware and software implementations:

- **Hardware Architecture:** The design will utilise the AD5941 SoC AFE, which offers a compact physical footprint and ultra-low power consumption for portable electrochemical and impedance-based sensing systems.
- **Microcontroller Integration:** The system will be driven by an ESP32-S3 microcontroller, chosen for its dual-core Xtensa LX7 architecture and native Wi-Fi capabilities.
- **Physical User Interface:** To solve the headless limitation of existing embedded designs, the hardware will include a standalone physical user interface consisting of an onboard display, two push buttons, and a rotary encoder.
- **Electrochemical Interface:** The system will interface with a standard three-electrode setup consisting of a working, reference, and counter electrode. This three electrode setup can be used with a SPE or a traditional dupont connector.
- **Software and Firmware:** Firmware will be developed in C. It will enable the system to execute DPV measurements, control the local GUI, and host an embedded web server for seamless data extraction via Wi-Fi.

1.4 Report Layout

This section discuesses the reports layout and the contents of each chapter. The report is structured into ....

Chapter 2

Literature

2.1 Introduction to Review

This chapter reviews the literature relevant to the design of a low-cost, high-performance potentiostat for Differential Pulse Voltammetry (DPV). The purpose of this review is to establish the electrochemical and hardware fundamentals required to design such an instrument, and to survey the existing landscape of commercial and open-source potentiostat designs in order to identify a gap in the current market. The project's aim is the development of a low-cost potentiostat with improved performance and usability relative to existing open-source alternatives.

2.2 Electrochemical Background

2.2.1 Basics of Electrochemical Cells

A potentiostat controls the potential between the reference and working electrode while simultaneously measuring the current flowing between the counter and the working electrode. This is a popular three-electrode setup for measurement, illustrated in Figure 2.1. The Working Electrode (WE) is the focal area of electrochemical reactions [1]. In a three-electrode system the potential around it is measured and manipulated to study the desired catalyst or analyte, where an analyte is a specific substance or chemical component that is the subject of analysis.

The Reference Electrode (RE) provides a stable, known potential that is used as a benchmark to control and monitor the working electrode's potential [2]. Current flow through the reference electrode is minimal to ensure that its potential remains accurate [2]. The Counter Electrode (CE) (also called the auxiliary electrode) completes the circuit, allowing current to be sourced or sunk between it and the working electrode, which prevents

the reference electrode from being affected [2]. The reaction at the counter electrode should be monitored, as corrosion at the electrode could interfere with measurement accuracy [3]. The three-electrode setup is the foundational hardware configuration for the voltammetric techniques that will be discussed later, namely Cyclic Voltammetry (CV), Linear Sweep Voltammetry (LSV), Square Wave Voltammetry (SWV), DPV, and stripping voltammetry.

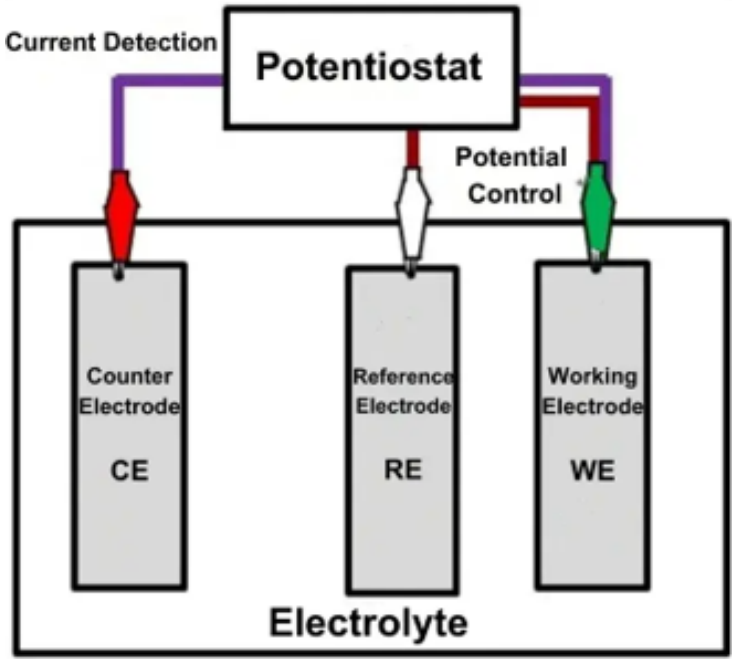


Figure 2.1: The three-electrode electrochemical cell configuration (working, reference and counter electrodes).

2.2.2 Voltammetry as a Family of Techniques (CV, LSV, SWV, DPV, Stripping Voltammetry)

In any voltammetric measurement, applying a voltage to an electrode generates two types of current:

- **Faradaic current:** The actual current generated by the transfer of electrons during the oxidation or reduction of the target analyte [4]. This is the desired signal.
- **Charging (capacitive) current:** The background current generated simply by charging the electrical double-layer at the electrode surface, which acts like a capacitor caused by the Helmholtz plane [4]. This causes background noise.

The same three-electrode setup handles each of the following techniques in a different way:

- **Linear Sweep (LSV) and Cyclic Voltammetry (CV):** The potentiostat applies a continuous, linear voltage ramp (or a triangular sweep for CV). The counter electrode continuously sinks or sources the resulting faradaic current.

- **Differential Pulse (DPV) and Square Wave Voltammetry (SWV):** Instead of a continuous ramp, the potentiostat applies a series of discrete voltage pulses or steps. By measuring the current at specific times during the pulse, the system can mathematically subtract the background capacitive (charging) current. This dramatically increases sensitivity, making DPV and SWV the preferred techniques for trace-level detection in biosensing applications.
- **Stripping voltammetry:** This is a two-step process that still relies on the exact same electrodes. First, the potentiostat holds a constant potential to electroplate (pre-concentrate) the analyte onto the working electrode. It then applies a sweep or pulse waveform (often an LSV, DPV, or SWV waveform) to "strip" the analyte back into the solution, yielding a large current spike.

2.2.3 Differential Pulse Voltammetry Theory

DPV superimposes small pulses of potential onto a linear potential ramp or staircase. The current difference is measured before and near the end of each pulse to reduce background charging current and achieve high sensitivity [5], although this comes at the cost of time resolution [6]. DPV builds on the foundations of linear sweep methods; however, it uses a more complex, discontinuous voltage waveform.

2.2.4 Waveform Description (Pulse Height, Pulse Width, Step Potential, Current Sampling Windows)

In DPV, the excitation waveform consists of a foundational staircase ramp with a discrete voltage pulse superimposed onto each step, as shown in Figure 2.2. The geometry of this waveform is defined by the pulse amplitude, which dictates the voltage magnitude of the superimposed pulse, and the pulse width, which specifies how long this elevated potential is held before the system returns to the baseline step potential.

Crucially, a potentiostat executing a DPV sweep does not measure current continuously. Instead, the analog front-end executes two precise, discrete current samples per cycle. The first sample (I_{before}) is recorded immediately before the pulse is applied, and the second sample (I_{end}) is captured at the end of the pulse duration. The final analytical output for that specific voltage step is derived by subtracting the baseline sample from the pulse sample, yielding a differential current

$$\Delta I = I_{\text{end}} - I_{\text{before}}. \tag{2.1}$$

This differential data point is what is ultimately plotted against the applied step potential to form the voltammogram.

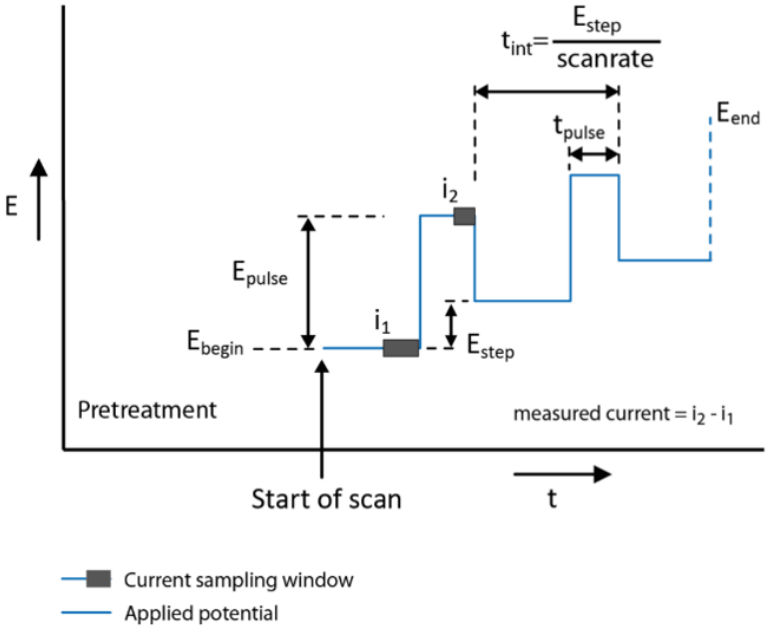


Figure 2.2: The DPV excitation waveform, showing the staircase ramp, superimposed pulses, and the two current-sampling instants (I_{before} and I_{end}) per step.

2.2.5 Why DPV Improves Sensitivity / Reduces Background (Capacitive) Current Compared to LSV/CV

When a changing voltage occurs (dV/dt) over a brief period, a large spike in capacitive charging current occurs at the electrode-electrolyte interface. In continuous voltage sweeps such as CV or LSV, this capacitive noise is always present. The capacitive charging current decays exponentially as $e^{-t/RC}$, whereas the faradaic current decays much more slowly and is governed by the Cottrell equation, $1/\sqrt{t}$. To combat this, the second current measurement is taken at the end of the pulse width; this allows the double-layer capacitor time to fully charge so that the capacitive noise has died out, leaving only the faradaic signal. The resulting measurement has a much higher Signal-to-Noise Ratio (SNR), allowing for the detection of much lower analyte concentrations. Analytes of interest span a large range of applications.

2.2.6 Typical Analytical Applications (Trace Metals, Pharmaceuticals, Biosensing)

Because DPV effectively filters out capacitive charging current, it achieves exceptional sensitivity and low limits of detection, making it a foundational technique across multiple analytical fields. In environmental monitoring, DPV is frequently used in tandem with anodic stripping methods to detect trace heavy metals, such as lead and arsenic, at parts-per-billion concentrations in water systems. Within the pharmaceutical industry, it is used

to analyse the purity of drug formulations and to study the oxidation mechanisms of active pharmaceutical ingredients. Furthermore, DPV is increasingly critical in the development of modern electrochemical biosensors. The biological fluids analysed in these systems (e.g. food or interstitial fluid) are highly resistive and generate significant background noise; by mathematically rejecting this noise, DPV allows microfluidic lab-on-a-chip devices to isolate the minute faradaic currents produced by specific biomarkers, enabling rapid and highly sensitive point-of-care diagnostics. All of these applications are controlled by a potentiostat.

2.3 Potentiostat Fundamentals

This section explores the fundamentals of a potentiostat, namely its role, design, performance parameters, and implementation approaches.

2.3.1 The Role of a Potentiostat

The role of a potentiostat is to control the control amplifier and the current-to-voltage amplifier (Transimpedance Amplifier (TIA)), and to buffer the reference electrode. For reference electrode buffering, the electrode must remain perfectly stable at a known half-cell potential; to prevent the voltage from drifting, zero current is allowed to flow through the electrode. This is achieved using an electrometer, an operational amplifier with an extremely high input impedance.

The control amplifier drives the system by constantly comparing the desired input voltage, set by a Digital-to-Analog Converter (DAC), with the actual voltage measured at the reference electrode. The resulting error signal is minimised by forcing current through the counter electrode and into the solution, which actively compensates for any changes in the electrochemical cell's resistance. The current-to-voltage converter (TIA) holds the working electrode at a 0 V reference (virtual ground) while catching every electron produced or consumed by the reaction. These electrons generate a small faradaic current that is converted into a potential, which can then be measured by an Analog-to-Digital Converter (ADC) [7]. Each of the three amplifiers is connected to one of the three electrodes.

2.3.2 The Classic 3-Operational Amplifier Potentiostat Topology

This section translates the three roles above into a physical circuit architecture, shown in Figure 2.3.

- **The feedback loop:** The 3-op-amp topology forms a negative feedback loop. The electrochemical cell itself (between the CE and RE) is physically placed inside the

feedback path of the control amplifier.

* **Op-amp A (the driver):** The control amplifier.
* **Op-amp B (the buffer):** The RE voltage follower.
* **Op-amp C (the sensor):** The TIA connected to the WE.

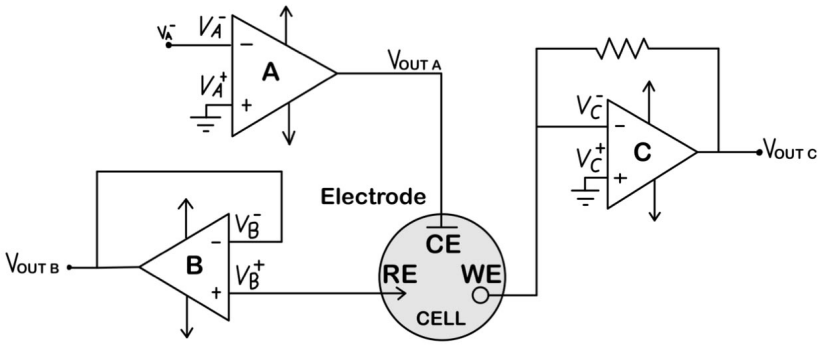


Figure 2.3: The classic 3-operational-amplifier potentiostat topology, showing the control amplifier, RE buffer, and TIA connected to the three electrodes.

2.3.3 Key Performance Parameters

This section establishes the criteria used to evaluate the effectiveness of the hardware, namely current measurement range and resolution, compliance voltage, noise floor, bandwidth, and technique support.

* **Current measurement range and resolution:** In microfluidic biosensing, very small faradaic signals are generated. The potentiostat must be able to accurately measure down to the nanoampere (nA) or picoampere (pA) range without clipping or saturating the signal.
* **Compliance voltage:** This is the maximum voltage the control amplifier can physically output. Compliance voltage describes the ability of a power supply to push a much higher maximum voltage in order to keep a steady current flowing through a high-resistance load. If a fluid sample has high electrical resistance, the control amplifier needs a high compliance voltage to push the necessary current through the counter electrode.
* **Noise floor and Limit of Detection (LOD):** The LOD of an electrochemical biosensor is fundamentally constrained by the inherent noise floor of the potentiostat's analog front end. In high-sensitivity applications, the TIA requires large feedback resistors to convert picoampere-level faradaic currents into measurable voltages.

However, these resistive components inherently generate thermal (Johnson–Nyquist) noise, which scales with resistance and temperature. Therefore, a microfluidic diagnostic device targeting trace biomarkers (generating nanoampere or picoampere signals) requires a potentiostat topology with an exceptionally low electrical noise floor. If the peak-to-peak amplitude of the circuit's electronic noise exceeds the faradaic signal generated by the biomarker, reliable detection is physically impossible.

* **Bandwidth / scan rate limits:** This is the maximum speed at which the potentiostat can alter the applied potential. Fast techniques such as SWV require high-bandwidth operational amplifiers to avoid signal distortion.

* **Technique support:** A system's ability to run CV, DPV, or Electrochemical Impedance Spectroscopy (EIS) depends on its DAC resolution and internal waveform generators [8].

2.3.4 Analog Front-End Approaches

Transitioning from the theoretical 3-op-amp topology to a physical implementation requires selecting an appropriate hardware architecture. While commercial benchtop potentiostats offer uncompromising analytical performance, their bulky form factors and high costs render them unsuitable for portable or point-of-care microfluidic diagnostics. Consequently, engineers developing custom electrochemical systems typically choose between two primary Analog Front-End (AFE) strategies: constructing a discrete circuit using individual operational amplifiers, or utilising a fully integrated System-on-Chip (SoC) AFE, such as the AD5941. Each approach presents distinct engineering trade-offs regarding power consumption, physical footprint, and architectural flexibility, which are summarised in Table 2.1.

Table 2.1: Comparison of discrete op-amp and integrated AFE IC implementation approaches.

Implementation Approach	Advantages	Trade-offs
Discrete op-amp design	Unlimited architectural flexibility; capable of massive compliance voltages; component-level tuning for extreme precision	Large physical PCB footprint; high power consumption; complex board-level noise routing; higher cost of materials
Integrated AFE ICs	Millimeter-scale physical footprint; ultra-low power consumption for Internet of Things (IoT); built-in hardware digital filtering; significantly lower per-unit cost	Hard-capped compliance voltage ($\leq$ 3 V); fixed internal current measurement ranges; black-box topology limits deep customisation; requires complex SPI/I2C firmware development

2.4 Survey of Existing Potentiostat Designs

The democratisation of electrochemical instrumentation has led to a rapid expansion of custom potentiostat designs. This section surveys the architectural evolution of these devices, establishing a performance baseline and identifying existing gaps in standalone usability and embedded intelligence.

2.4.1 Commercial Instruments

Commercial systems, such as the PalmSens EmStat or Metrohm Autolab, serve as the gold standard for electrochemical analysis. They offer uncompromising precision, massive dynamic ranges (pA to mA), rigorous factory calibration, and robust proprietary software suites. These instruments fall short in that they are cost-prohibitive (often exceeding thousands of dollars), physically bulky, and operate as closed-source devices, making them impossible to integrate deeply into custom, low-cost microfluidic or Internet of Things (IoT) architectures. Nevertheless, they are exceedingly valuable in that they provide the foundational performance benchmarks against which all open-source and custom hardware must be validated [9].

2.4.2 Early Open-Source Designs

The inception of open-source electrochemistry was marked by devices like the CheapStat [10]. These devices proved that rudimentary analytical electrochemistry could be achieved with a Bill of Materials (BOM) under $100, dramatically lowering the barrier to entry for educational and resource-limited settings. Since these first-generation potentiostats were

built on early 8-bit microcontrollers, they suffered from poor ADC/DAC resolution, narrow voltage compliance windows, and a lack of wireless connectivity. CheapStat pioneered the open-hardware movement in electrochemistry, establishing the viability of DIY discrete op-amp topologies [10].

2.4.3 General-Purpose / High-Performance Open-Source Designs

To bridge the performance gap between early DIY builds and commercial units, high-performance open-source designs such as the DStat [11] were developed. The DStat drastically improved noise performance, offered multi-technique support (CV, SWV, DPV, CA), and provided an excellent Python-based Graphical User Interface (GUI). Unfortunately, it remains tethered to a PC via USB for power and data processing; while highly accurate, this reliance on an external computer negates true field portability. This was the first time that custom discrete analog front-ends could achieve near-commercial sensitivity limits [11].

2.4.4 Portable / Wireless / Smartphone-Integrated Designs

The development of Bluetooth Low Energy (BLE) led to a generation of smartphone-integrated devices, such as the UWED and various ESP32-based We-VoltamoStat iterations [12]. These devices achieved true field portability by offloading the processing and display requirements onto the user's smartphone, reducing the hardware footprint of the potentiostat itself. However, these designs are locked into their companion application: if the proprietary application is not maintained to support operating system updates, the hardware becomes obsolete [12]. Furthermore, Bluetooth pairing can be unreliable in electrically noisy industrial environments [13].

2.4.5 EIS-Capable / AFE-Chip-Based Designs

The introduction of highly integrated, mixed-signal AFEs catalysed a shift away from bulky discrete op-amp circuits [14]. Systems like ABE-Stat, HunStat (2024), and HunStat2 (2025) characterise this generation. Utilising SoCs such as the AD5941 enables these devices to support complex, high-frequency techniques like EIS while consuming fractions of a milliwatt; they boast exceptionally low electronic noise floors and millimeter-scale footprints [15, 16]. The development of ESP32 microcontrollers further aided performance as well as the appeal for DIY projects. However, the firmware required to configure the AFE registers is exceptionally complex, and devices such as HunStat are typically deployed as data-loggers; they lack built-in user interfaces and rely on raw serial data dumps for analysis. These two factors are an unfortunate downside of these systems [16].

2.4.6 Self-Hosted Web-Interface / Onboard-Display Designs

To solve the challenge of being locked into a companion app, recent ESP32-based potentiostats have integrated embedded web servers. NanoStat [17] hosts an HTML5/JavaScript server, meaning any Wi-Fi-enabled device can control it via a standard browser. DM-Stat [18] and SaxStat [19] follow similar "OpenEChem" philosophies. A major benefit of this approach is ultimate cross-platform compatibility (Windows, macOS, iOS, Android) without requiring the user to install any software; however, hosting a web server consumes significant Microcontroller Unit (MCU) RAM and flash memory [17].

There is a potential novelty gap in including a physical onboard display in such a system. A review of these systems reveals that they are almost exclusively headless. This highlights a distinct gap in the literature: a dual-interface approach would allow for rapid, standalone field testing without requiring the user to reach for a smartphone, while still preserving web-based data extraction capabilities.

2.4.7 Machine-Learning-Assisted Voltammetry

The most recent paradigm shift involves embedding Machine Learning (ML) algorithms directly onto the potentiostat's microcontroller. Examples include embedded multi-layer perceptrons for heavy-metal stripping voltammetry and models classifying DPV responses for Dengue/Zika biomarkers on highly constrained hardware ($<$10 kB RAM) [20, 21]. Such models shift the system from mere data *collection* to on-device data *interpretation*, allowing the device to output actionable results (e.g. "positive/negative" or specific concentrations) rather than forcing a human expert to interpret a noisy voltammogram. These systems are less attractive in that they require extensive prior data collection to train the models, and inference algorithms consume heavy compute cycles on standard MCUs.

Table 2.2 summarises the surveyed designs against the proposed architecture for this project.

Table 2.2: Comparison of surveyed potentiostat designs against the proposed architecture.

Design / Platform	**MCU / AFE Architecture**	**Primary Techniques Supported**	**Connectivity & Interface**	**Onboard Display?**	**ML / Onboard Analysis?**	**Open-Source Completeness**
Commercial (e.g. EmStat)	Proprietary	CV, DPV, SWV, EIS, CA	USB / Proprietary PC GUI	No	No (PC-based)	Closed source
CheapStat (2011)	Atmel AVR / Discrete	CV, SWV, CA	None / Basic LCD output	Yes (Basic LCD)	No	Full (PCB/Firmware)
DStat (2015)	Atmel XMEGA / Discrete	CV, DPV, SWV, CA	USB / Python PC GUI	No	No	Full (PCB/Firmware/GUI)
Smartphone-Integrated (e.g. UWED)	BLE MCU / Discrete	CV, DPV, SWV	Bluetooth / Custom App	No	No	Partial (Hardware focus)
HunStat2 (2025)	Mixed / AD5941 SoC	CV, DPV, SWV, EIS	USB / Serial Datalogging	No	No	Full (Hardware/Firmware)
NanoStat (2022)	ESP32 / Discrete	CV, SWV, CA	Wi-Fi / Embedded Web GUI	No (Headless)	No	Full (Hardware/Web GUI)
Proposed ML Architecture	ESP32 / AD5941 SoC	CV, DPV, SWV	Wi-Fi Web GUI + Standalone	Yes	Yes (Edge ML)	TBD

Chapter 3

System Modeling & Theoretical Design

3.1 System Architecture

The proposed hardware architecture of the potentiostat is illustrated in Figure 3.1 is divided into high-level software logic and discrete hardware components. The architecture is designed to be modular, allowing for easy integration of various components. The analog front-end serves as the core of the system, providing the necessary tools for electrochemical measurements. The microcontroller manages data acquisition, processing, and communication. The physical user interface allows for local control and user input and control. The three-electrode setup ensures accurate electrochemical measurements, making the system suitable for detecting trace levels of antiretroviral medications in water samples.

The software architecture manages the analytical core and connectivity of the system. It is driven by real-time operating system and features an embedded web server and Wi-Fi capabilities for data extraction and review alongside data storage. The analytical core manages the DPV waveform generation and differential current calculations, while a local GUI and display driver ensure the physical interface is operational without the need for an external smart device.

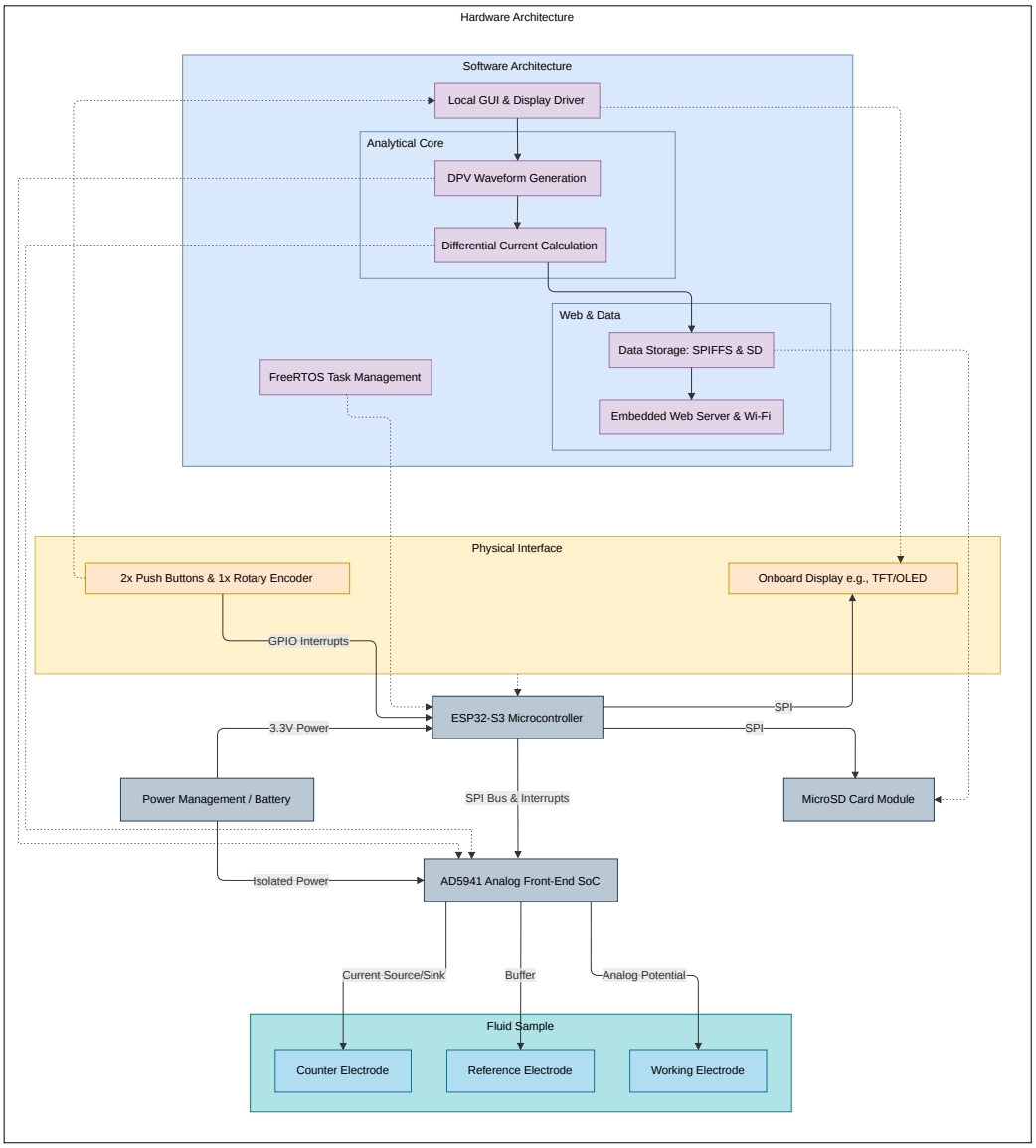


Figure 3.1: System Architecture Diagram

3.2 Electrochemical Cell & Equivalent Circuit Modeling

It is possible to model the electrochemical cell as a Randle's equivalent circuit, as seen in Figure 3.2. After defining the system architecture required to interface with the electrochemical cell, it is required to mathematically model what the cell presents to the system. The Randle equivalent circuit is a complex non-ideal impedance model that changes depending on frequency, applied potential and the electrochemical reaction occurring at the working electrode. To design a control loop capable of driving this load reliably, the cell must first be reduced to an equivalent electrical model. A widely used

representation of the cell is the Randles equivalent circuit [22]. This equivalent circuit models the electrode and electrolyte interface as a series solution resistance (R_s) in series with a parallel combination of the double-layer capacitance (C_{dl}) and the charge-transfer resistance (R_{ct}). The Warburg impedance (Z_W) is included to account for diffusion effects at low frequencies. This model allows for the analysis of the electrochemical cell's behavior under different conditions and is essential for designing the potentiostat's control loop [23].

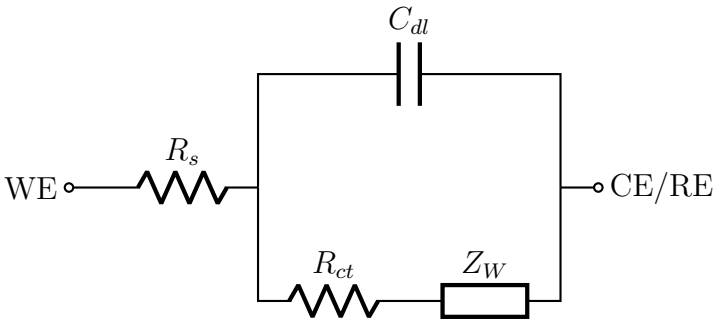


Figure 3.2: Randles equivalent circuit modeling the electrochemical cell interface.

The link between DPV waveforms and the Randles equivalent circuit is found through identifying the electrical demand it places on the potentiostat. The combination of the R_s and R_{ct} resistors is the total resistive load that must be driven by the control amplifier to sustain the target faradaic current. This resistive load dictates the compliance voltage that the system requires [24]. The combination of R_s and C_{dl} form a RC time constant which dictates how quickly the double layer capacitance charges after each step in potential. This time constant governs how long a DPV pulse should be held to ensure that the background current has sufficiently decayed relative to the faradaic current [7]. Rather than being arbitrary selections, parameters such as pulse width, TIA gain, and control-amplifier bandwidth are strictly dictated by the expected R_s, C_{dl}, and R_{ct} values of the target analyte-electrode interface. If the sampling scheme or amplifier bandwidth fails to match this characteristic RC time constant, the system cannot effectively filter out the unwanted capacitive component of the measured current[23].

3.2.1 Proving Capacitive Current Rejection in DPV

DPV's effectiveness relies on the different decay rates of the two current components generated at an electrode following a potential step.

3.2.1.1 Total Current at the Electrode

When a potential step is applied, the total measured current is the sum of the faradaic and capacitive components:

$$I_{total}(t) = I_f(t) + I_c(t) \tag{3.1}$$

Current Type	Source	Decay Function	Decay Rate
Capacitive (I_c)	Charging the electrical double layer	$e^{-t/RC}$	Very fast (exponential)
Faradaic (I_f)	Electron transfer	$1/\sqrt{t}$	Slower (inverse square root)

Table 3.1: Comparison of Capacitive and Faradaic Currents

The capacitive current models the electrode-solution interface as an RC circuit, decaying exponentially [25]:

$$I_c(t) = \frac{\Delta E}{R_s}\, e^{-t/R_s C_{dl}} \tag{3.2}$$

Where:

* R_s is the solution resistance.
* C_{dl} is the double-layer capacitance.
* ΔE is the applied potential step.

The faradaic current is governed by the Cottrell equation for planar diffusion, decaying with the inverse square root of time [7]:

$$I_f(t) = \frac{nFA\sqrt{D}\, C^*}{\sqrt{\pi t}} \tag{3.3}$$

Where:

* n is the number of electrons transferred.
* F is Faraday's constant.
* A is the electrode area.
* D is the diffusion coefficient.
* C^* is the bulk analyte concentration.

Because an exponential function ($e^{-t/RC}$) tends toward zero faster than an inverse square root function ($1/\sqrt{t}$), by waiting for a sufficient amount of time after applying the pulse ensures the capacitive current disappeared while the faradaic current remains.

3.2.1.2 The Pulse-Sampling Scheme

To utilise the property difference between the two current types, DPV applies a series of short voltage pulses superimposed on a linear staircase ramp. To isolate the faradaic current, the potentiostat samples the current twice per pulse cycle:

1. I_{before}: sampled just before the pulse is applied.
2. I_{end}: sampled at the very end of the pulse duration ($t = \tau_{pulse}$).

3.2.1.3 The Differential Proof

The final output plotted in a DPV voltammogram is the difference between these two samples:

$$\Delta I = I_{end} - I_{before} \tag{3.4}$$

Expanding using the component currents:

$$\Delta I = (I_{f,end} + I_{c,end}) - (I_{f,before} + I_{c,before}) \tag{3.5}$$

Finding a solution to the decay mathematics to solve the problem:

* Since I_{before} is sampled just before the pulse, the system has been at a baseline potential for a sufficient amount of time so that any previous capacitive transient current has decayed completely. Thus, $I_{c,before} \approx 0$.
* Since I_{end} is sampled at the very end of the pulse duration, the new capacitive transient caused by the pulse has also decayed to near zero. Thus, $I_{c,end} \approx 0$.

Therefore, the differential in Equation 3.5 simplifies to:

$$\Delta I \approx I_{f,end} - I_{f,before} \tag{3.6}$$

By subtracting I_{before} from I_{end}, DPV mathematically rejects the background charging current, leaving only the faradaic signal of interest.

3.2.1.4 Why the Output Forms a Peak

Because ΔI represents the change in faradaic current for a small, discrete step in potential (ΔE), the resulting plot of ΔI versus E is effectively an approximation of the derivative of a standard current-potential curve:

$$\Delta I \approx \frac{dI_f}{dE} \cdot \Delta E \tag{3.7}$$

For a reversible system (electron transfer is so fast the system is always in thermodynamic equilibrium) governed by the Nernst equation, taking the derivative of the standard sigmoidal voltammogram produces a Gaussian-like peak. This peak reaches its maximum amplitude exactly at the half-wave potential ($E_{1/2}$), providing an incredibly sensitive and easily quantifiable signal [26].

3.3 Analog Front-End Stability & Control Loop Modeling

The generalized transfer function for the transimpedance amplifier is defined as:

$$H(s) = \frac{V_{\text{out}}(s)}{I_{\text{in}}(s)} = -\frac{R_f}{1 + sR_fC_f} \tag{3.8}$$

3.3.1 Impact of C_{DL} on TIA Loop Stability

The working electrode connects directly to the summing junction (virtual ground) of the transimpedance amplifier (TIA). Because the double-layer capacitance, C_{dl}, physically exists between the working electrode and the bulk solution, it appears electrically as a shunt capacitance from the TIA's inverting input node to ground – functionally identical to the junction capacitance of a photodiode in a conventional TIA design [**?**]. This section derives the TIA transfer function to show how C_{dl} introduces an additional pole into the feedback loop, degrading phase margin and threatening stability.

3.3.1.1 Circuit Model and Node Equation

Let R_f be the TIA feedback resistor, C_{dl} the shunt capacitance at the inverting node, and $A(s)$ the open-loop gain of the operational amplifier. Let V_n be the voltage at the inverting node and I_{in} the faradaic current sourced by the electrochemical reaction. Applying KCL at the summing node:

$$I_{in} = V_n\, sC_{dl} + \frac{V_n - V_{out}}{R_f} \tag{3.9}$$

Since the op-amp enforces $V_{out} = -A(s)V_n$ (with the non-inverting input grounded), we have:

$$V_n = -\frac{V_{out}}{A(s)} \tag{3.10}$$

Substituting Equation (3.10) into Equation (3.9) and solving for V_{out}/I_{in} gives the closed-loop transimpedance:

$$Z_T(s) \equiv \frac{V_{out}}{I_{in}} = \frac{-R_f}{1 + \dfrac{1 + sR_fC_{dl}}{A(s)}} \tag{3.11}$$

3.3.1.2 Noise Gain and the Loop Gain Pole

Define the *noise gain* of the feedback network as:

$$NG(s) = 1 + sR_fC_{dl} \tag{3.12}$$

Equation (3.12) shows that C_{dl} introduces a *zero* into the noise gain at:

$$f_z = \frac{1}{2\pi R_fC_{dl}} \tag{3.13}$$

Above f_z, the noise gain rises at $+20\,\text{dB/decade}$. The op-amp itself is modelled as a single-dominant-pole system, $A(s) = \omega_t/s$ near the crossover region, where $\omega_t = 2\pi\,\text{GBW}$ is the unity-gain angular frequency. The loop gain is therefore:

$$T(s) = \frac{A(s)}{NG(s)} = \frac{\omega_t}{s(1 + sR_fC_{dl})} \tag{3.14}$$

Equation (3.14) reveals the core stability problem: the zero that C_{dl} introduced into the noise gain (Eq. (3.12)) becomes an additional *pole* in the loop gain, located at:

$$\omega_{p2} = \frac{1}{R_fC_{dl}} \tag{3.15}$$

Combined with the op-amp's intrinsic pole at the origin (from the integrator-like ω_t/s term), the loop gain $T(s)$ now contains *two* poles below the crossover frequency.

3.3.1.3 Phase Margin Degradation

The closed-loop crossover frequency ω_{cl}, where $|T(j\omega_{cl})| = 1$, is found by setting the magnitude of Equation (3.14) to unity. For $\omega_{cl} \gg 1/(R_fC_{dl})$ (the typical high-bandwidth case):

$$|T(j\omega)| \approx \frac{\omega_t}{\omega^2 R_fC_{dl}} = 1 \quad \Longrightarrow \quad \omega_{cl} = \sqrt{\frac{\omega_t}{R_fC_{dl}}} \tag{3.16}$$

At this crossover frequency, each of the two poles in $T(s)$ contributes close to $-90°$ of phase shift, for a combined phase approaching $-180°$. Since a negative feedback loop becomes unstable once the total loop phase reaches $-180°$ at the frequency where $|T(j\omega)| = 1$, the phase margin,

$$\phi_{PM} = 180° - |\angle T(j\omega_{cl})|\,, \tag{3.17}$$

collapses toward 0°. In practice this manifests as pronounced ringing, overshoot, or sustained oscillation in the measured current whenever the electrode's C_{dl} is large relative to $1/(\omega_t R_f)$ – precisely the condition encountered with high-sensitivity potentiostat designs, where R_f must be large (to resolve nA–pA faradaic currents) and C_{dl} can range from nanofarads to microfarads depending on electrode area and surface roughness [**?**].

3.3.1.4 Compensation

Stability is restored by adding a small feedback capacitor C_f in parallel with R_f, introducing a compensating pole into the noise gain *before* the C_{dl}-induced zero dominates. Choosing C_f such that the noise gain pole occurs at the geometric mean of f_z and the amplifier's gain-bandwidth product restores a single-pole roll-off at crossover, yielding a phase margin close to 45°–65° [**?**]:

$$C_f = \sqrt{\frac{C_{dl}}{2\pi R_f \,\mathrm{GBW}}} \tag{3.18}$$

Equation (3.18) demonstrates that the compensation capacitor required for stable operation is a direct function of the electrode's double-layer capacitance, reinforcing that the electrochemical cell – modelled via the Randles equivalent circuit – is not merely a passive measurement target but an active determinant of the analog front-end's stability margin.

3.4 Voltammetric Digital Signal Processing & Peak Extraction

The differential current is calculated as:

$$\Delta I = I_{\text{end}} - I_{\text{before}} \tag{3.19}$$

The exact peak potential is automatically isolated by solving for the root of the derivative:

$$\frac{d(\Delta I)}{dE} = 0 \tag{3.20}$$

3.5 Embedded Communication Protocol & GUI State Machine

Figure 3.3: Finite state machine diagram for managing asynchronous tasks and WebSocket communication.

Chapter 4

Detailed Hardware & Software Implementation

This chapter details the transition from theoretical modeling to physical realization. It justifies the silicon selection, outlines the two-layer printed circuit board (PCB) design and routing strategies, and breaks down the embedded C++ architecture required to drive the analog front-end (AFE) and serve the real-time web interface.

4.1 Choosing a Microcontroller

In the previous section the requirements of the features that need to be available on the microcontroller were explored. The AD5941 interface requires a fast, dedicated hardware SPI bus and a few GPIO pins for interrupts/resets. The web server requires an integrated Wi-Fi radio, a robust TCP/IP stack, and significant RAM/flash to store and serve HTML/JavaScript files. The onboard display requires either I2C (for simple OLEDs) or a high-speed SPI/parallel interface for colour TFT screens. The edge-ML engine requires relatively high clock speeds, hardware floating-point support, and enough memory to store model weights (likely using TensorFlow Lite for Microcontrollers).

4.1.1 Choosing the Microcontroller Family

The range of microcontroller options is vast; there are three main architecture options available: STM32, Nordic (nRF52), and ESP, which are compared in Table 4.1.

Table 4.1: Comparison of candidate microcontroller families.

Platform	Advantage	Disadvantage
STM32	Golden standard for microcontrollers	Additional hardware needed for Wi-Fi
Nordic (nRF52)	Dominates the BLE space	Additional hardware needed for Wi-Fi
ESP	Native Wi-Fi, Bluetooth	—

Since ESP is the only family of microprocessors with native Wi-Fi, Bluetooth, and sufficient processing power, a decision needs to be made regarding which variant of these chips to select. Table 4.2 compares the relevant ESP32 variants.

Table 4.2: Comparison of candidate ESP32 variants.

Variant	Core Architecture	AI/ML Hardware Acceleration	Best Fit For
ESP32 (Classic)	Dual-core Xtensa LX6	No	General purpose, older designs
ESP32-C3	Single-core RISC-V	No	Ultra-low power, but too weak for ML and Web
ESP32-S3	Dual-core Xtensa LX7	Yes (Vector Instructions)	Heavy compute, displays, ML

4.2 Physical Circuitry & Electrode Interface Design

Figure 4.1: Custom two-layer PCB layout highlighting ground plane stitching and analog/digital signal separation.

4.3 Analog Front-End (AFE) Register Configuration

Listing 4.1: Example AD5941 AFE sequencer initialization for a voltammetric sweep.

```
// TODO: Insert your actual AD5941 sequencer or register initialization c
void AD5941_Init(void) {
```

```
    // 1. Wake up AFE
    AD5941_WakeUp();
    // 2. Configure High-Speed DAC
    // 3. Set PGA gain and Sinc filters
}
```

4.4 ESP32-S3 Firmware Architecture & Real-Time Loop

Listing 4.2: FreeRTOS task pinning to separate cores on the ESP32-S3.

```
// TODO: Insert your FreeRTOS xTaskCreatePinnedToCore logic here
xTaskCreatePinnedToCore(
    DSP_SPI_Task,        // Task function
    "DSP_Task",          // Task name
    8192,                // Stack size
    NULL,                // Parameters
    configMAX_PRIORITIES - 1, // High priority
    &Task1,              // Task handle
    0                    // Pinned to Core 0
);

xTaskCreatePinnedToCore(
    WebSocket_UI_Task,
    "UI_Task",
    8192,
    NULL,
    1,                   // Lower priority
    &Task2,
    1                    // Pinned to Core 1
);
```

4.5 Wireless Web GUI & Interface Realization

Figure 4.2: Real-time web interface served from SPIFFS, plotting DPV scans via WebSockets.

Chapter 5

Summary and Conclusions

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Appendix A

Appendix title goes here